ENGT 320 Applied Thermal Systems
Quiz 5 Formula Sheet

$$m = \rho \forall$$
$$\dot{\forall} = \frac{\forall}{t}$$
$$q = mc_p \Delta T$$
$$q = -kA\frac{dT}{dt}$$
$$q = hA\left(T_s - T_\infty\right)$$
$$q = \varepsilon \sigma A\left(T_s^4 - T_{sur}^4\right)$$
$$\sigma = 5.67 \times 10^{-8} \frac{W}{m^2 K^4}$$
$$\Delta E_{sys} = \Sigma E_{in} - \Sigma E_{out}$$
$$\frac{dE_{sys}}{dt} = \Sigma \dot{E}_{in} - \Sigma \dot{E}_{out}$$
$$q'' = \frac{q}{A}$$
$$\hat{q}'' = -k_x \frac{\partial T}{\partial x}\hat{i} - k_y \frac{\partial T}{\partial y}\hat{j} - k_z \frac{\partial T}{\partial z}\hat{k}$$
$$k = \frac{9\gamma - 5}{4} \frac{c_v}{\pi d^2} \sqrt{\frac{M_w k_B T}{N_A \pi}}$$
$$k_B = 1.381 \times 10^{-23} \frac{J}{K}$$
$$N_A = 6.022 \times 10^{23}$$
$$\alpha = \frac{k}{\rho c_p}$$
$$q_{x+dx} = q_x + \frac{\partial q_x}{\partial x} dx$$
$$q_{y+dy} = q_y + \frac{\partial q_y}{\partial y} dy$$
$$q_{z+dz} = q_z + \frac{\partial q_z}{\partial z} dz$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = 0$$

$$\frac{\partial}{\partial x}\left(k\frac{\partial T}{\partial x}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{\partial}{\partial y}\left(k\frac{\partial T}{\partial y}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial^2 T}{\partial \phi^2} + \frac{\partial^2 T}{\partial z^2} + \frac{\dot{q}}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = 0$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{1}{r^2}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{\partial}{\partial z}\left(k\frac{\partial T}{\partial z}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(k\sin\theta\frac{\partial T}{\partial \theta}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2 T}{\partial \phi^2} + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\frac{\partial T}{\partial \theta}\right) + \frac{\dot{q}}{k} = \frac{1}{\alpha}\frac{\partial T}{\partial t}$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(k\sin\theta\frac{\partial T}{\partial \theta}\right) + \dot{q} = 0$$

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(kr^2\frac{\partial T}{\partial r}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{1}{r^2\sin^2\theta}\frac{\partial}{\partial \phi}\left(k\frac{\partial T}{\partial \phi}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$\frac{1}{r^2\sin\theta}\frac{\partial}{\partial \theta}\left(k\sin\theta\frac{\partial T}{\partial \theta}\right) + \dot{q} = \rho c_p \frac{\partial T}{\partial t}$$
$$T(0,t) = T_s$$
$$-k\frac{\partial T}{\partial x}\Big|_{x=0} = q_s''$$
$$\frac{\partial T}{\partial x}\Big|_{x=0} = 0$$
$$-k\frac{\partial T}{\partial x}\Big|_{x=0} = h\left(T_\infty - T(0,t)\right)$$

$$C_1 = \frac{T_{s,2} - T_{s,1}}{L}$$
$$C_2 = T_{s,1}$$
$$T(x) = \left(T_{s,2} - T_{s,1}\right)\frac{x}{L} + T_{s,1}$$
$$q = \frac{kA}{L}\left(T_{s,1} - T_{s,2}\right)$$
$$R_{t,cond} = \frac{T_{s,1} - T_{s,2}}{q} = \frac{L}{kA}$$
$$R = \frac{V}{I}$$
$$R_{t,conv} = \frac{T_s - T_\infty}{q} = \frac{1}{hA}$$
$$R_{t,rad} = \frac{T_s - T_{sur}}{q} = \frac{1}{h_r A}$$
$$h_r = \varepsilon\sigma\left(T_s + T_{sur}\right)\left(T_s^2 + T_{sur}^2\right)$$
$$q = UA\Delta T$$
$$U = \frac{1}{R_{tot}A}$$
$$R_{tot} = \Sigma R_t = \frac{1}{UA}$$

$$R_{t,c}'' = \frac{T_A - T_B}{q''}$$
$$q = \frac{k_{eff}A}{L}\left(T_1 - T_2\right)$$
$$k_{eff,min} = \frac{1}{\frac{1-\varepsilon}{k_s} + \frac{\varepsilon}{k_f}}$$
$$k_{eff,max} = \varepsilon k_f + (1-\varepsilon)k_s$$
$$R_{t,cond} = \frac{\ln\frac{r_2}{r_1}}{2\pi kL}$$
$$R_{t,conv} = \frac{1}{h2\pi rL}$$

$$\frac{d^2T}{dx^2} + \left(\frac{1}{A_c}\frac{dA_c}{dx}\right)\frac{dT}{dx} - \left(\frac{1}{A_c}\frac{h}{k}\frac{dA_s}{dx}\right)\left(T - T_\infty\right) = 0$$

$$A_s = Px$$
$$\frac{d^2T}{dx^2} - \left(\frac{hP}{kA_c}\right)\left(T - T_\infty\right)$$
$$\frac{d^2\theta}{dx^2} - m^2\theta = 0$$
$$\theta = T(x) - T_\infty$$
$$m^2 = \frac{hP}{kA_c}$$
$$P = 2w + 2t$$
$$A_c = wt$$
$$P = \pi D$$
$$A_c = \frac{\pi}{4}D^2$$
$$\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$$
$$\theta_b = T_b - T_\infty$$
$$M = \sqrt{hPKA_c}\theta_b$$
$$\varepsilon_f = \frac{q_f}{hA_{c,b}\theta_b}$$
$$\varepsilon_f = \sqrt{\frac{kP}{hA_c}}$$
$$R_{t,f} = \frac{\theta_b}{q_f}$$
$$R_{t,b} = \frac{1}{hA_{c,b}}$$
$$\varepsilon_f = \frac{R_{t,b}}{R_{t,f}}$$
$$\eta_f = \frac{q_f}{q_{max}} = \frac{q_f}{hA_f\theta_b}$$
$$\eta_o = \frac{q_t}{q_{max}} = \frac{q_t}{hA_t\theta_b}$$
$$A_t = NA_f + A_b$$
$$q_t = N\eta_f hA_f\theta_b + hA_b\theta_b$$
$$\eta_o = 1 - \frac{NA_f}{A_t}\left(1 - \eta_f\right)$$
$$R_{t,o} = \frac{\theta_b}{q_t} = \frac{1}{\eta_o hA_t}$$

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$$\frac{\partial^2 T}{\partial x^2}\Big|_{m,n} \approx \frac{\frac{\partial T}{\partial x}\Big|_{m+\frac{1}{2},n} - \frac{\partial T}{\partial x}\Big|_{m-\frac{1}{2},n}}{\Delta x}$$

$$\frac{\partial T}{\partial x}\Big|_{m+\frac{1}{2},n} \approx \frac{T\Big|_{m+1,n} - T\Big|_{m,n}}{\Delta x}$$

$$\frac{\partial T}{\partial x}\Big|_{m-\frac{1}{2},n} \approx \frac{T\Big|_{m,n} - T\Big|_{m-1,n}}{\Delta x}$$

$$\frac{\partial^2 T}{\partial x^2}\Big|_{m,n} \approx \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{(\Delta x)^2}$$

$$\frac{\partial^2 T}{\partial y^2}\Big|_{m,n} \approx \frac{T_{m+1,n} + T_{m-1,n} - 2T_{m,n}}{(\Delta y)^2}$$

$$\sum_{i=1}^{4} q_{(i)\to(m,n)} + \dot{q}\,(\Delta x \cdot \Delta y \cdot 1) = 0$$

$$q_{(m-1,n)\to(m,n)} = k\,(\Delta y \cdot 1)\,\frac{T_{m-1,n} - T_{m,n}}{\Delta x}$$

$$q_{(m+1,n)\to(m,n)} = k\,(\Delta y \cdot 1)\,\frac{T_{m+1,n} - T_{m,n}}{\Delta x}$$

$$q_{(m,n+1)\to(m,n)} = k\,(\Delta x \cdot 1)\,\frac{T_{m,n+1} - T_{m,n}}{\Delta y}$$

$$q_{(m,n-1)\to(m,n)} = k\,(\Delta x \cdot 1)\,\frac{T_{m,n-1} - T_{m,n}}{\Delta y}$$

$$T_{m,n+1} + T_{m,n-1} + T_{m+1,n} + T_{m-1,n} + \frac{\dot{q}\cdot 1\cdot(\Delta x)^2}{k} - 4T_{m,n} = 0$$

$$AT = C$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1N} \\ a_{21} & a_{22} & \cdots & a_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ a_{N1} & a_{N2} & \cdots & a_{NN} \end{bmatrix}$$

$$T = \begin{bmatrix} T_1 \\ T_2 \\ \vdots \\ T_N \end{bmatrix}$$

$$C = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_N \end{bmatrix}$$

$$T = A^{-1}C$$

$$\frac{u(y)}{u_\infty} = 0.99$$

$$C_f = \frac{\tau_s}{\rho u_\infty^2/2}$$

$$\tau_s = \mu \frac{\partial u}{\partial y}\Big|_{y=0}$$

$$\frac{T_s - T(y)}{T_s - T_\infty} = 0.99$$

$$q_s'' = -k\frac{\partial T}{\partial y}\Big|_{y=0}$$

$$q_s'' = h\,(T_s - T_\infty)$$

$$h = \frac{-k\frac{\partial T}{\partial y}\Big|_{y=0}}{T_s - T_\infty}$$

$$q = \int_{A_s} q''\,dA_s = (T_s - T_\infty)\int_{A_s} h\,dA_s$$

$$\overline{q}'' = \overline{h}\,(T_s - T_\infty)$$

$$\overline{q} = \overline{q}''\,A_s = \overline{h}A_s\,(T_s - T_\infty)$$

$$\overline{h} = \frac{1}{A_s}\int_{A_s} h\,dA_s$$

$$\overline{h} = \frac{1}{L}\int_0^L h\,dx$$

$$Re_x = \frac{\rho u_\infty x}{\mu}$$

$$x^* = \frac{x}{L}$$

$$y^* = \frac{y}{L}$$

$$u^* = \frac{u}{V}$$

$$\mathrm{v}^* = \frac{\mathrm{V}}{V}$$

$$T^* = \frac{T - T_s}{T_\infty - T_s}$$

$$p^* = \frac{p_\infty}{\rho V^2}$$

$$u^*\frac{\partial u^*}{\partial x^*} + \mathrm{v}^*\frac{\partial u^*}{\partial y^*} = -\frac{dp^*}{dx^*} + \frac{1}{Re_L}\frac{\partial^2 u^*}{\partial y^{*2}}$$

$$Re_L = \frac{VL}{\nu}$$

$$\nu = \frac{\mu}{\rho}$$

$$u^*\frac{\partial T^*}{\partial x^*} + \mathrm{v}^*\frac{\partial T^*}{\partial y^*} = \frac{1}{Re_L Pr}\frac{\partial^2 T^*}{\partial y^{*2}}$$

$$Pr = \frac{\nu}{\alpha}$$

$$Nu = \frac{hL}{k} = \frac{\partial T^*}{\partial y^*}\Big|_{y^*=0}$$

$$\overline{Nu} = \frac{\overline{h}L}{k}$$

$$T_f = \frac{T_s + T_\infty}{2}$$

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Isothermal Flat Plate

$$\frac{\delta}{\delta_t} \approx Pr^{1/3}$$

Laminar:

$$\delta = \frac{5}{\sqrt{\frac{u_\infty}{\nu x}}} = \frac{5x}{\sqrt{Re_x}}$$

$$C_{f,x} = \frac{\tau_{s,x}}{\frac{\rho u_\infty^2}{2}} = 0.664 Re_x^{-1/2}$$

$$Nu_x = \frac{h_x x}{k} = 0.332 Re_x^{1/2} Pr^{1/3}$$

$$(Pr \gtrsim 0.6)$$

$$\overline{C}_{f,x} = 1.328 Re_x^{-1/2}$$

$$\overline{Nu}_x = \frac{\overline{h}_x x}{k} = 0.664 Re_x^{1/2} Pr^{1/3}$$

$$(Pr \gtrsim 0.6)$$

Turbulent:

$$\delta = 0.37 x Re_x^{-1/2}$$

$$C_{f,x} = 0.0592 Re_x^{-1/5}$$

$$\left(Re_{x,c} \lesssim Re_x \lesssim 10^8\right)$$

$$Nu_x = 0.0296 Re_x^{4/5} Pr^{1/3}$$

$$(0.6 \lesssim Pr \lesssim 60)$$

$$A = 0.037 Re_{x,c}^{4/5} - 0.664 Re_{x,c}^{1/2}$$

$$\overline{C}_{f,L} = 0.074 Re_L^{-1/5} - \frac{2A}{Re_L}$$

$$\left(Re_{x,c} \lesssim Re_L \lesssim 10^8\right)$$

$$\overline{Nu}_L = \left(0.037 Re_L^{4/5} - A\right) Pr^{1/3}$$

$$(0.6 \lesssim Pr \lesssim 60)$$

$$\left(Re_{x,c} \lesssim Re_L \lesssim 10^8\right)$$

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$$Re_D = \frac{\rho V D}{\mu} = \frac{VD}{\nu}$$

$$C_D = \frac{F_D}{A_f\left(\frac{\rho V^2}{2}\right)}$$

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Cylinder

$$\overline{Nu}_D = \frac{\overline{h}D}{k} = C Re_D^m Pr^{1/3}$$

$$(Pr \gtrsim 0.7)$$

See Table 5

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Various Geometries

$\overline{Nu}_D = \frac{\overline{h}D}{k} = CRe_D^m Pr^{1/3}$

$(Pr \gtrsim 0.7)$

See Table 6

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Sphere

$A_s = \pi D^2$

$C_D = \frac{25}{Re_D}$

$(Re_D \lesssim 0.5)$

$\overline{Nu}_D = 2 + \left(0.4Re_D^{1/2} + 0.06Re_D^{2/3}\right) Pr^{0.4} \left(\frac{\mu}{\mu_s}\right)^{1/4}$

$(0.71 \lesssim Pr \lesssim 380)$

$\left(3.5 \lesssim Re_D \lesssim 7.6 \times 10^4\right)$

$\left(1.0 \lesssim \frac{\mu}{\mu_s} \lesssim 3.2\right)$

. . . . . .

$m = \rho u_m A_c$

$Re_D = \frac{\rho u_m D}{\mu} = \frac{u_m D}{\nu}$

$\left(\frac{x_{fd,t}}{D}\right) \approx 0.05 Re_D Pr$

$f = \frac{-\frac{dP}{dx} D}{\rho \frac{u_m^2}{2}}$

$f = \frac{64}{Re_D}$

$\frac{1}{\sqrt{f}} = -2.0 \log\left[\frac{e/D}{3.7} + \frac{2.51}{Re_D\sqrt{f}}\right]$

$dq_{conv} = mc_p\left[(T_m + dT_m) - T_m\right] = mc_p dT_m$

$dq_{conv} = q_s^{''} P dx$

$\frac{dT_m}{dx} = \frac{q_s^{''} P}{mc_p} = \frac{P}{mc_p} h\left(T_s - T_m\right)$

$T_m(x) = T_{m,i} + \frac{q_s^{''} P}{mc_p} x$

$\Delta T = T_s - T_m$

$\frac{dT_m}{dx} = -\frac{d(\Delta T)}{dx} = \frac{P}{mc_p} h \Delta T$

$\frac{T_s - T_m(x)}{T_s - T_{m,i}} = e^{-\frac{Px}{mc_p}\overline{h}}$

$\Delta T_{lm} = \frac{\Delta T_o - \Delta T_i}{\ln\left(\frac{\Delta T_o}{\Delta T_i}\right)}$

$\Delta T_o = T_s - T_{m,o}$

$\Delta T_i = T_s - T_{m,i}$

$q_{conv} = \overline{h} A_s \Delta T_{lm}$

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Circular Tubes

Fully dev. lam. w/ $q^{''} = \mathbb{C}$:

$Nu_D = 4.36$

Fully dev. lam. w/ $T_s = \mathbb{C}$:

$Nu_D = 3.66$

Fully dev. turb. w/ small ΔT:

$Nu_D = 0.023 Re_D^{4/5} Pr^n$

$n = 0.4, \quad T_s > T_m$

$n = 0.3, \quad T_s < T_m$

$(0.6 \lesssim Pr \lesssim 160)$

$(Re_D \gtrsim 10,000)$

$(L/D \gtrsim 10)$

Fully dev. turb. w/ large ΔT:

$Nu_D = 0.027 Re_D^{4/5} Pr^{1/3} \left(\frac{\mu}{\mu_s}\right)^{0.14}$

$(0.7 \lesssim Pr \lesssim 16,700)$

$(Re_D \gtrsim 10,000)$

$(L/D \gtrsim 10)$

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Noncircular Tubes

$D_h = \frac{4A_c}{P}$

$Nu_D = \frac{hD_h}{k}$

See Table 7

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$\frac{1}{UA} = \frac{1}{U_c A_c} = \frac{1}{U_h A_h} = \frac{1}{(hA)_c} + R_w + \frac{1}{(hA)_h}$

$\frac{1}{UA} = \frac{1}{(\eta_o hA)_c} + \frac{R_{f,c}^{''}}{(\eta_o A)_c} + R_w + \frac{R_{f,h}^{''}}{(\eta_o A)_h} + \frac{1}{(\eta_o hA)_h}$

$q = m_h\left(i_{h,i} - i_{h,o}\right)$

$q = m_c\left(i_{c,i} - i_{c,o}\right)$

$q = m_h c_{p,h}\left(T_{h,i} - T_{h,o}\right)$

$q = m_c c_{p,c}\left(T_{c,o} - T_{c,i}\right)$

$q = UA\Delta T_m$

$q = UA\Delta T_{lm}$

$\Delta T_{lm} = \frac{\Delta T_2 - \Delta T_1}{\ln\frac{\Delta T_2}{\Delta T_1}} = \frac{\Delta T_1 - \Delta T_2}{\ln\frac{\Delta T_1}{\Delta T_2}}$

$\Delta T_1 = T_{h,1} - T_{c,1} = T_{h,i} - T_{c,i}$

$\Delta T_2 = T_{h,2} - T_{c,2} = T_{h,o} - T_{c,o}$

$\Delta T_1 = T_{h,1} - T_{c,1} = T_{h,i} - T_{c,o}$

$\Delta T_2 = T_{h,2} - T_{c,2} = T_{h,o} - T_{c,i}$

$q_{max} = C_{min}\left(T_{h,i} - T_{c,i}\right)$

$\varepsilon = \frac{C_h(T_{h,i} - T_{h,o})}{C_{min}(T_{h,i} - T_{c,i})}$

$\varepsilon = \frac{C_c(T_{c,o} - T_{c,i})}{C_{min}(T_{h,i} - T_{c,i})}$

$q = \varepsilon C_{min}\left(T_{h,i} - T_{c,i}\right)$

$NTU = \frac{UA}{C_{min}}$

$\lambda = \frac{c}{\nu}$

$c_o = 2.998 \times 10^8 \frac{m}{s}$

$E = \varepsilon\sigma T^4$

$J = E + \rho G$

$q_{rad}^{''} = J - G$

$q_{rad}^{''} = \varepsilon\sigma T^4 - \alpha G$

$G = G_{ref} + G_{abs} + G_{tr}$

$\rho + \alpha + \tau = 1$

$\rho + \alpha = 1$

$d\omega = \frac{dA_n}{r^2}$

$dA_n = r^2 \sin\theta d\theta d\phi$

$d\omega = \sin\theta d\theta d\phi$

$\omega_{hemi} = \int_0^{2\pi}\int_0^{\pi/2} \sin\theta d\theta d\phi = 2\pi sr$

$I_{\lambda,e}(\lambda,\theta,\phi) = \frac{dq}{(dA_1\cos\theta)d\omega d\lambda}$

$dq_\lambda = \frac{dq}{d\lambda} = I_{\lambda,e}(\lambda,\theta,\phi)\, dA_1 \cos\theta d\omega$

$dq_\lambda^{''} = I_{\lambda,e}(\lambda,\theta,\phi)\cos\theta d\omega = I_{\lambda,e}(\lambda,\theta,\phi)\cos\theta\sin\theta d\omega$

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$$E_\lambda(\lambda) = \int_0^{2\pi}\int_0^{\pi/2} I_{\lambda,e}(\lambda,\theta,\phi)\cos\theta\sin\theta d\theta d\phi$$

$$E = \int_0^\infty E_\lambda(\lambda)\,d\lambda$$

$$E_\lambda(\lambda) = \pi I_{\lambda,e}(\lambda)$$

$$E = \pi I_e$$

$$G_\lambda(\lambda) = \int_0^{2\pi}\int_0^{\pi/2} I_{\lambda,i}(\lambda,\theta,\phi)\cos\theta\sin\theta d\theta d\phi$$

$$G = \int_0^\infty G_\lambda(\lambda)\,d\lambda$$

$$J_\lambda(\lambda) = \int_0^{2\pi}\int_0^{\pi/2} I_{\lambda,e+r}(\lambda,\theta,\phi)\cos\theta\sin\theta d\theta d\phi$$

$$J = \int_0^\infty J_\lambda(\lambda)\,d\lambda$$

$$E_{\lambda,b}(\lambda,T) = \pi I_{\lambda,T}(\lambda,T) = \frac{C_1}{\lambda^5\left[e^{(C_2/\lambda T)}-1\right]}$$

$$C_1 = 3.742\times10^8 W\mu m^4/m^2$$

$$C_2 = 1.439\times10^4 \mu m K$$

$$\lambda_{max}T = C_3 = 2898\mu m K$$

$$E_b = \pi I_b = \int_0^\infty E_{\lambda,b}d\lambda = \sigma T^4$$

$$F_{(\lambda_1-\lambda_2)} = F_{0-\lambda_2} - F_{0-\lambda_1} = \frac{\int_0^{\lambda_2} E_{\lambda,b}d\lambda - \int_0^{\lambda_1} E_{\lambda,b}d\lambda}{\sigma T^4}$$

$$F_{(0-\lambda)} = \frac{\int_0^{\lambda} E_{\lambda,b}d\lambda}{\sigma T^4}$$

$$F_{ij} = \frac{1}{A_i}\int_{A_i}\int_{A_j}\frac{\cos\theta_i\cos\theta_j}{\pi R^2}dA_j dA_i$$

$$F_{ji} = \frac{1}{A_j}\int_{A_i}\int_{A_j}\frac{\cos\theta_i\cos\theta_j}{\pi R^2}dA_j dA_i$$

$$A_i F_{ij} = A_j F_{ji}$$

$$\sum_{j=1}^N F_{ij} = 1$$

$$J_i = E_{bi}$$

$$q_{ij} = q_{i\to j} - q_{j\to i}$$

$$q_{ij} = A_i F_{ij} E_{bi} - A_j F_{ji} E_{bj}$$

$$q_{ij} = A_i F_{ij}\sigma\left(T_i^4 - T_j^4\right)$$

$$q_i = \sum_{j=1}^N A_i F_{ij}\sigma\left(T_i^4 - T_j^4\right)$$

$$q_i = A_i(J_i - G_i)$$

$$q_i = A_i(E_i - \alpha_i G_i)$$

$$q_i = \frac{E_{bi}-J_i}{(1-\varepsilon_i)/\varepsilon_i A_i}$$

$$q_i = \sum_{j=1}^N A_i F_{ij}(J_i - J_j) = \sum_{j=1}^N \frac{J_i - J_j}{(A_i F_{ij})^{-1}}$$

$$q_1 = -q_2 = q_{12} = \frac{\sigma\left(T_1^4 - T_2^4\right)}{\frac{1-\varepsilon_1}{\varepsilon_1 A_1} + \frac{1}{A_1 F_{12}} + \frac{1-\varepsilon_2}{\varepsilon_2 A_2}}$$

$$q_1 = -q_2 = \frac{\sigma\left(T_1^4 - T_2^4\right)}{\frac{1-\varepsilon_1}{\varepsilon_1 A_1} + \frac{1}{A_1 F_{12} + \left[(1/A_1 F_{1R}) + (1/A_2 F_{2R})\right]^{-1}} + \frac{1-\varepsilon_1}{\varepsilon_1 A_1}}$$

$$\frac{J_1 - J_R}{1/A_1 F_{1R}} = \frac{J_R - J_2}{1/A_2 F_{2R}}$$

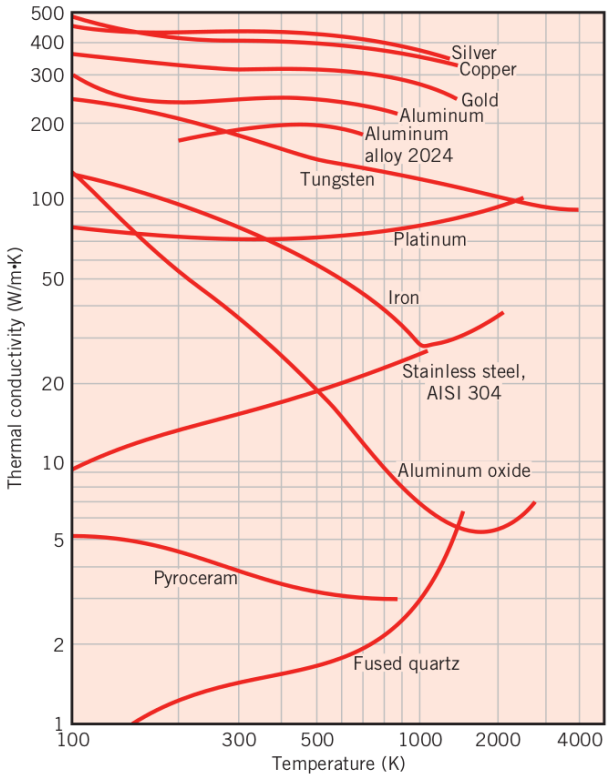
$$T_R = \left(\frac{J_R}{\sigma}\right)^{1/4}$$

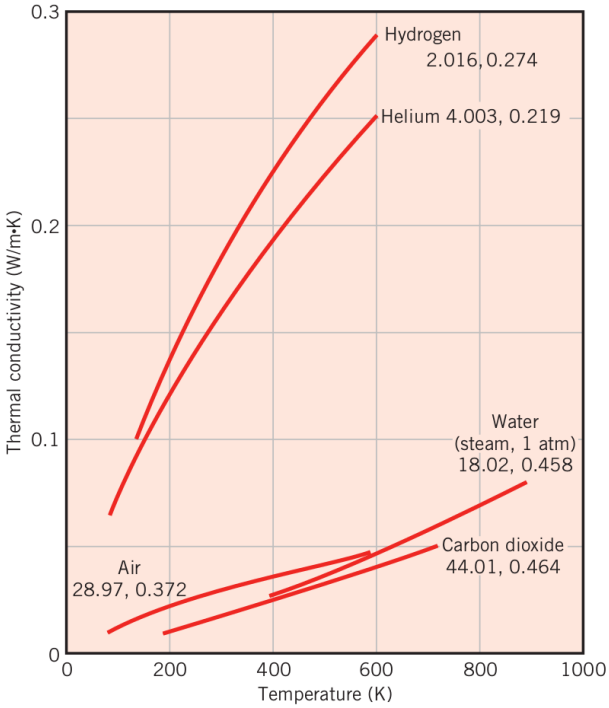
$$q_{i,ext} = q_{i,rad} + q_{i,conv} + q_{i,cond}$$

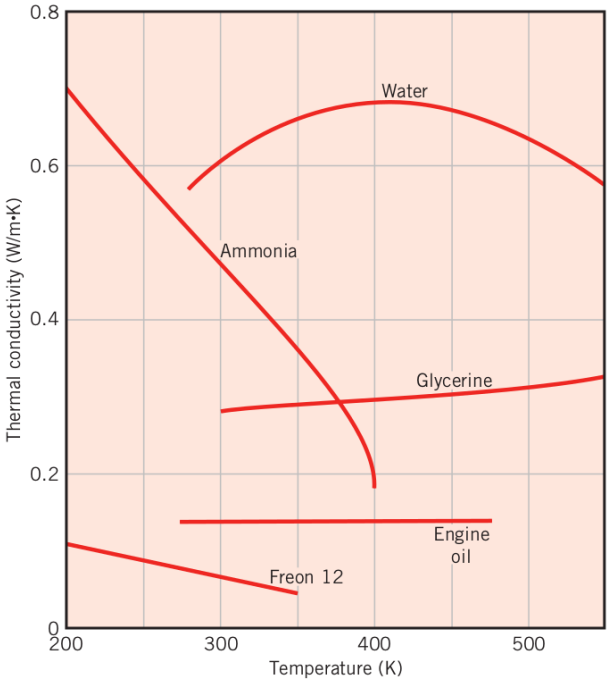
Lecture Note 14 formulas not needed

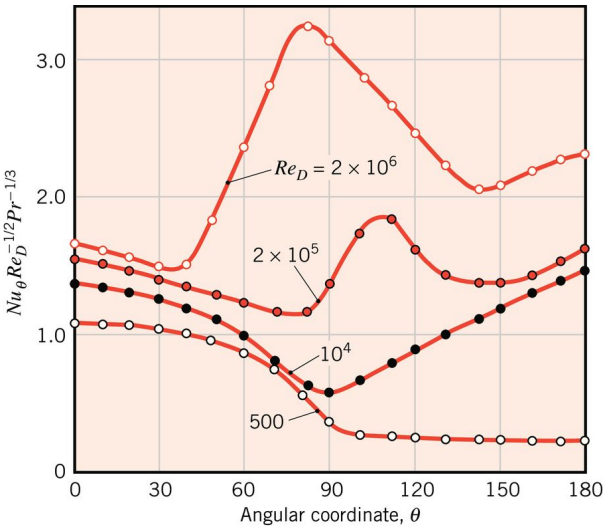
$$F_{ij} = \frac{q_{i\to j}}{A_i J_i}$$

$$dq_{i\to j} = I_i\cos\theta_i dA_i d\omega_{j-i} = J_i\frac{\cos\theta_i\cos\theta_j}{\pi R^2}dA_j dA_i$$

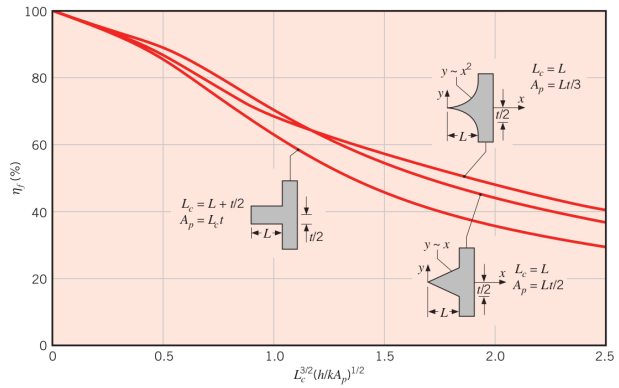

Thermal conductivity (W/m•K)
Temperature (K)
Silver
Copper
Gold
Aluminum
Aluminum alloy 2024
Tungsten
Platinum
Iron
Stainless steel, AISI 304
Aluminum oxide
Pyroceram
Fused quartz

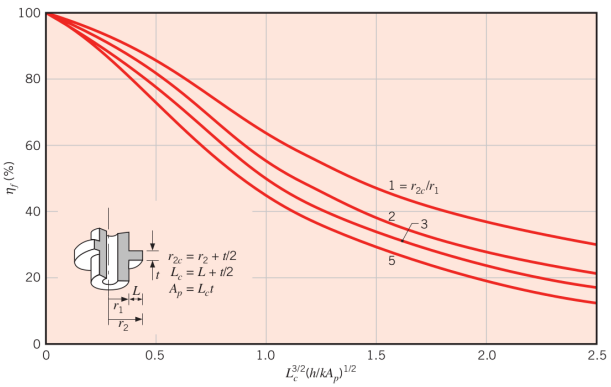

Thermal conductivity (W/m•K)
Temperature (K)
Hydrogen
2.016, 0.274
Helium 4.003, 0.219
Water
(steam, 1 atm)
18.02, 0.458
Carbon dioxide
44.01, 0.464
Air
28.97, 0.372


Thermal conductivity (W/m•K)
Temperature (K)
Water
Ammonia
Glycerine
Engine oil
Freon 12


$Nu_\theta Re_D^{-1/2} Pr^{-1/3}$
Angular coordinate, θ
$Re_D = 2 \times 10^6$
2×10^5
10^4
500

ENGT 320 Applied Thermal Systems
Quiz 5 Formula Sheet





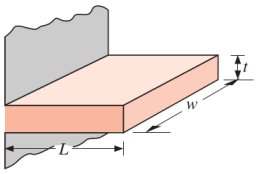
Straight Fins

Rectangular

$A_f = 2wL_c$

$L_c = L + (t/2)$

$A_p = tL$

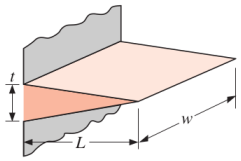


$$\eta_f = \frac{\tanh mL_c}{mL_c}$$

Triangular

$A_f = 2w[L^2 + (t/2)^2]^{1/2}$

$A_p = (t/2)L$



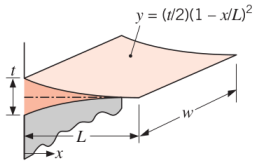
$$\eta_f = \frac{1}{mL}\frac{I_1(2mL)}{I_0(2mL)}$$

Parabolic

$A_f = w[C_1L + (L^2/t)\ln(t/L + C_1)]$

$C_1 = [1 + (t/L)^2]^{1/2}$

$A_p = (t/3)L$



$$\eta_f = \frac{2}{[4(mL)^2 + 1]^{1/2} + 1}$$

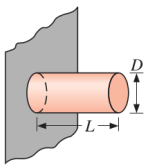
Pin Fins

Rectangular

$A_f = \pi DL_c$

$L_c = L + (D/4)$

$V = (\pi D^2/4)L$

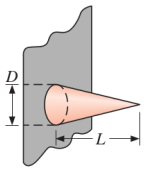


$$\eta_f = \frac{\tanh mL_c}{mL_c}$$

Triangular

$A_f = \frac{\pi D}{2}[L^2 + (D/2)^2]^{1/2}$

$V = (\pi/12)D^2L$



$$\eta_f = \frac{2}{mL}\frac{I_2(2mL)}{I_1(2mL)}$$

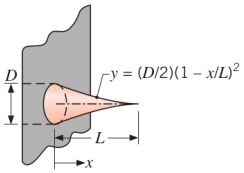
Parabolic

$A_f = \frac{\pi L^3}{8D}\{C_3C_4 - \frac{L}{2D}\ln[(2DC_4/L) + C_3]\}$

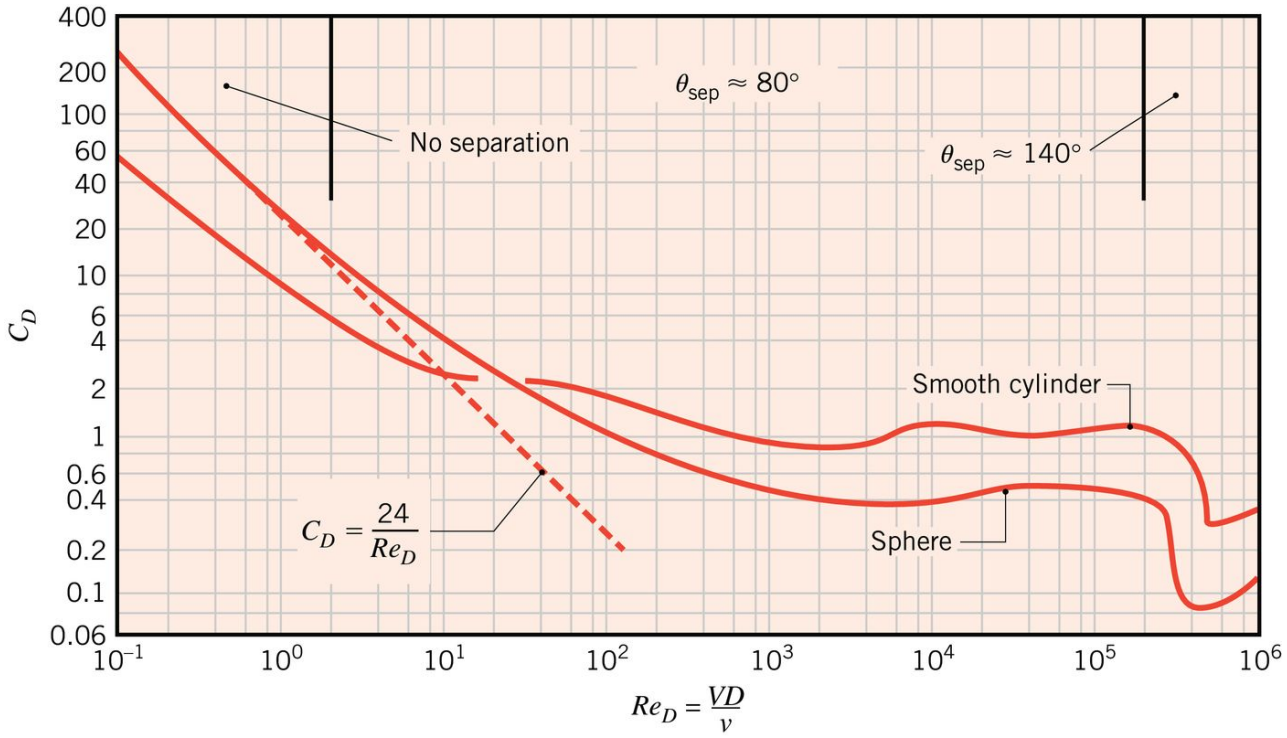
$C_3 = 1 + 2(D/L)^2$

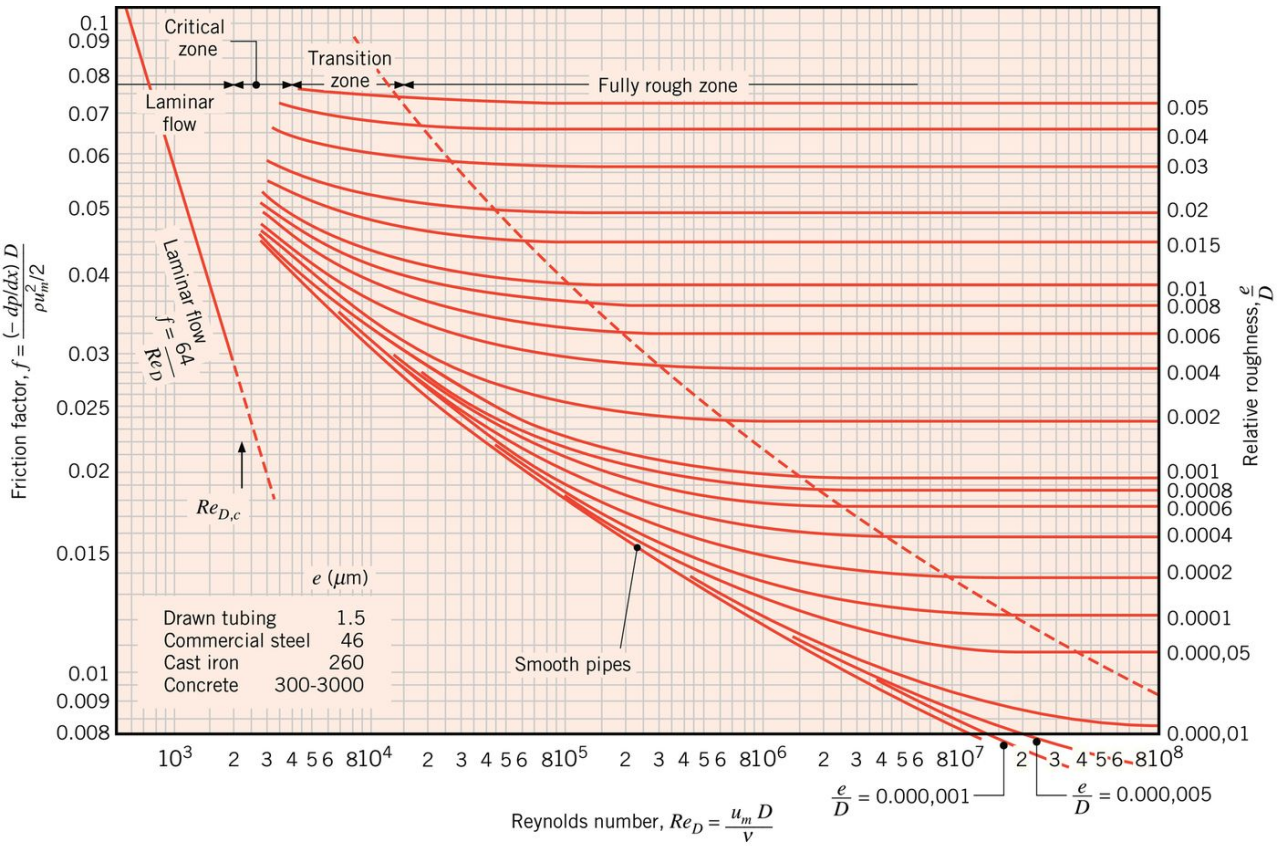
$C_4 = [1 + (D/L)^2]^{1/2}$

$V = (\pi/20)D^2L$



$$\eta_f = \frac{2}{[4/9(mL)^2 + 1]^{1/2} + 1}$$


C_D
400
200
100
60
40
20
10
6
4
2
1
0.6
0.4
0.2
0.1
0.06
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
$Re_D = \frac{VD}{\nu}$
No separation
$\theta_{sep} \approx 80°$
$\theta_{sep} \approx 140°$
Smooth cylinder
Sphere
$C_D = \frac{24}{Re_D}$

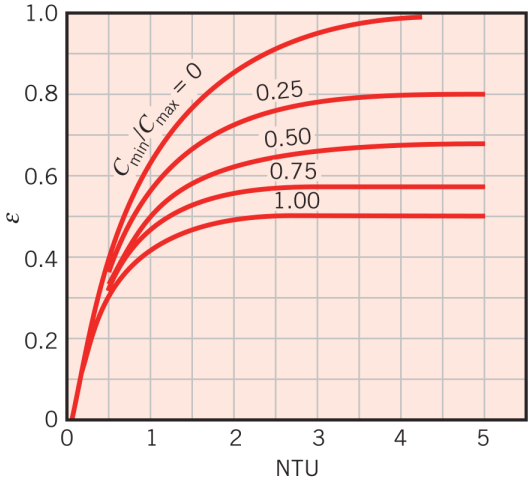

Friction factor, $f = \frac{(-dp/dx)\,D}{\rho u_m^2/2}$
Relative roughness, $\frac{e}{D}$
Critical zone
Transition zone
Fully rough zone
Laminar flow
Laminar flow $f = \frac{64}{Re_D}$
$Re_{D,c}$
e (μm)
Drawn tubing 1.5
Commercial steel 46
Cast iron 260
Concrete 300-3000
Smooth pipes
0.1
0.09
0.08
0.07
0.06
0.05
0.04
0.03
0.025
0.02
0.015
0.01
0.009
0.008
0.05
0.04
0.03
0.02
0.015
0.01
0.008
0.006
0.004
0.002
0.001
0.0008
0.0006
0.0004
0.0002
0.0001
0.000,05
0.000,01
10^3 2 3 4 5 6 8 10^4 2 3 4 5 6 8 10^5 2 3 4 5 6 8 10^6 2 3 4 5 6 8 10^7 2 3 4 5 6 8 10^8
Reynolds number, $Re_D = \frac{u_m D}{\nu}$
$\frac{e}{D} = 0.000,001$
$\frac{e}{D} = 0.000,005$

Flow Arrangement	Relation
Parallel ow	$\varepsilon = \dfrac{1 - \exp[-\text{NTU}(1 + C_r)]}{1 + C_r}$
Counterow	$\varepsilon = \dfrac{1 - \exp[-\text{NTU}(1 - C_r)]}{1 - C_r \exp[-\text{NTU}(1 - C_r)]} \quad (C_r < 1)$
	$\varepsilon = \dfrac{\text{NTU}}{1 + \text{NTU}} \quad (C_r = 1)$
Shell-and-tube	
One shell pass (2, 4, . . . tube passes)	$\varepsilon_1 = 2\left\{1 + C_r + (1 + C_r^2)^{1/2} \times \dfrac{1 + \exp[-(\text{NTU})_1(1 + C_r^2)^{1/2}]}{1 - \exp[-(\text{NTU})_1(1 + C_r^2)^{1/2}]}\right\}^{-1}$
n shell passes ($2n$, $4n$, . . . tube passes)	$\varepsilon = \left[\left(\dfrac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1}\right)^n - 1\right]\left[\left(\dfrac{1 - \varepsilon_1 C_r}{1 - \varepsilon_1}\right)^n - C_r\right]^{-1}$
Cross-ow (single pass)	
Both fluids unmixed	$\varepsilon = 1 - \exp\left[\left(\dfrac{1}{C_r}\right)(\text{NTU})^{0.22}\{\exp[-C_r(\text{NTU})^{0.78}] - 1\}\right]$
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$\varepsilon = \left(\dfrac{1}{C_r}\right)(1 - \exp\{-C_r[1 - \exp(-\text{NTU})]\})$
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$\varepsilon = 1 - \exp(-C_r^{-1}\{1 - \exp[-C_r(\text{NTU})]\})$
All exchangers ($C_r = 0$)	$\varepsilon = 1 - \exp(-\text{NTU})$

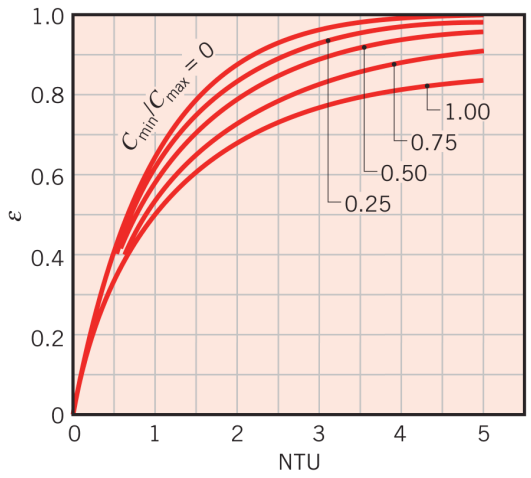
Flow Arrangement	Relation
Parallel ow	$\text{NTU} = -\dfrac{\ln[1 - \varepsilon(1 + C_r)]}{1 + C_r}$
Counterow	$\text{NTU} = \dfrac{1}{C_r - 1}\ln\left(\dfrac{\varepsilon - 1}{\varepsilon C_r - 1}\right) \quad (C_r < 1)$
	$\text{NTU} = \dfrac{\varepsilon}{1 - \varepsilon} \quad (C_r = 1)$
Shell-and-tube	
One shell pass (2, 4, . . . tube passes)	$(\text{NTU})_1 = -(1 + C_r^2)^{-1/2}\ln\left(\dfrac{E - 1}{E + 1}\right)$ $E = \dfrac{2/\varepsilon_1 - (1 + C_r)}{(1 + C_r^2)^{1/2}}$
n shell passes ($2n$, $4n$, . . . tube passes)	Use Equations 11.30b and 11.30c with $\varepsilon_1 = \dfrac{F - 1}{F - C_r} \quad F = \left(\dfrac{\varepsilon C_r - 1}{\varepsilon - 1}\right)^{1/n} \quad \text{NTU} = n(\text{NTU})_1$
Cross-ow (single pass)	
$C_{\max}$ (mixed), $C_{\min}$ (unmixed)	$\text{NTU} = -\ln\left[1 + \left(\dfrac{1}{C_r}\right)\ln(1 - \varepsilon C_r)\right]$
$C_{\min}$ (mixed), $C_{\max}$ (unmixed)	$\text{NTU} = -\left(\dfrac{1}{C_r}\right)\ln[C_r \ln(1 - \varepsilon) + 1]$
All exchangers ($C_r = 0$)	$\text{NTU} = -\ln(1 - \varepsilon)$

ENGT 320 Applied Thermal Systems
Quiz 5 Formula Sheet

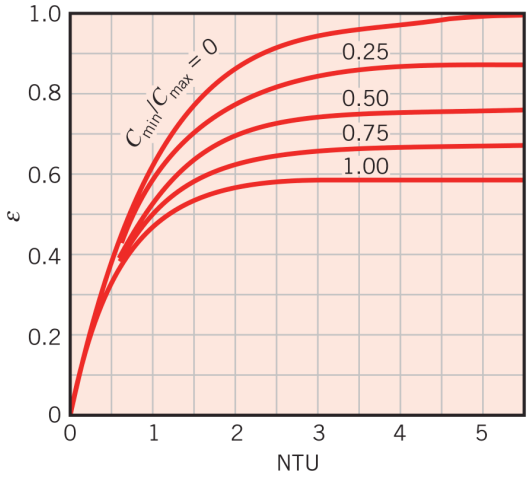
Parallel Flow



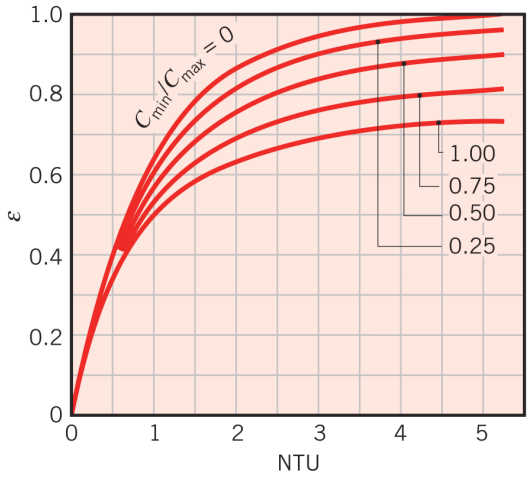
Counter Flow



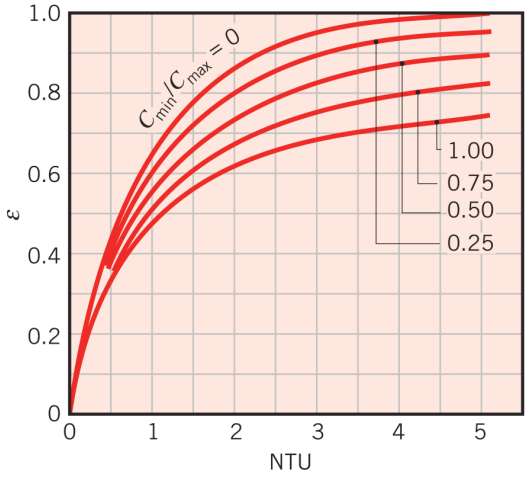
Shell and Tube - One Shell



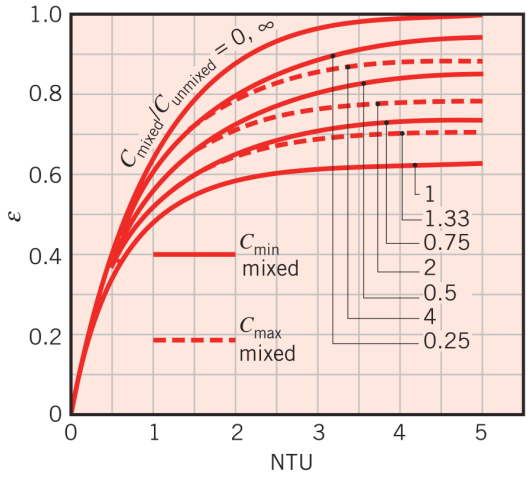
Shell and Tube - 2 Shells



Cross-flow Both Unmixed

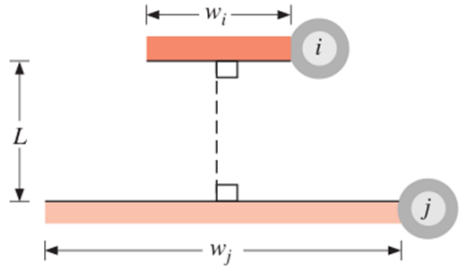


Cross-flow One Mixed/One Unmixed



Geometry	Relation

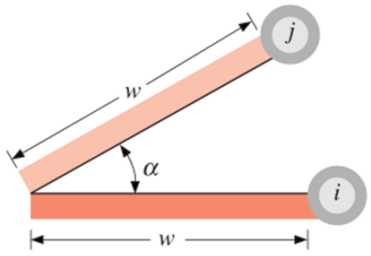
Parallel Plates with Midlines Connected by Perpendicular



$$F_{ij} = \frac{[(W_i + W_j)^2 + 4]^{1/2} - [(W_j - W_i)^2 + 4]^{1/2}}{2W_i}$$

$$W_i = w_i/L,\ W_j = w_j/L$$

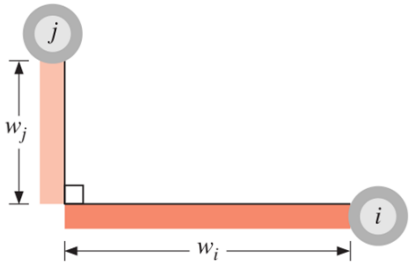
Inclined Parallel Plates of Equal Width and a Common Edge



$$F_{ij} = 1 - \sin\left(\frac{\alpha}{2}\right)$$

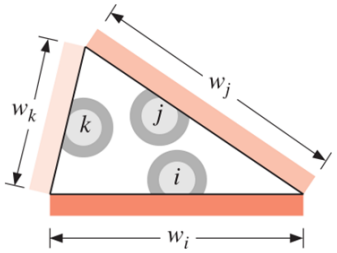
Geometry	Relation

Perpendicular Plates with a Common Edge



$$F_{ij} = \frac{1 + (w_j/w_i) - [1 + (w_j/w_i)^2]^{1/2}}{2}$$

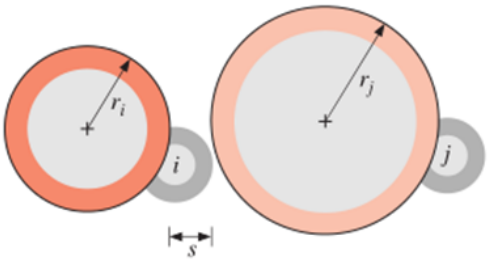
Three-Sided Enclosure



$$F_{ij} = \frac{w_i + w_j - w_k}{2w_i}$$

Geometry	Relation

Parallel Cylinders of Different Radii

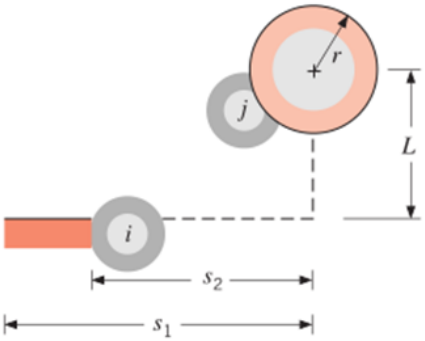


$$F_{ij} = \frac{1}{2\pi}\left\{\pi + [C^2 - (R+1)^2]^{1/2} - [C^2 - (R-1)^2]^{1/2} + (R-1)\cos^{-1}\left[\left(\frac{R}{C}\right) - \left(\frac{1}{C}\right)\right] - (R+1)\cos^{-1}\left[\left(\frac{R}{C}\right) + \left(\frac{1}{C}\right)\right]\right\}$$

$$R = r_j/r_i,\ S = s/r_i$$

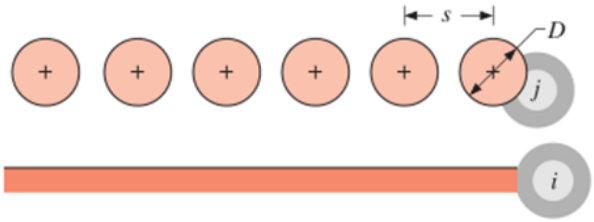
$$C = 1 + R + S$$

Cylinder and Parallel Rectangle

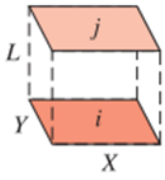
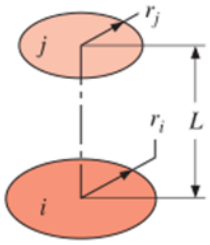
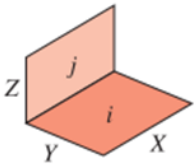


$$F_{ij} = \frac{r}{s_1 - s_2}\left[\tan^{-1}\frac{s_1}{L} - \tan^{-1}\frac{s_2}{L}\right]$$

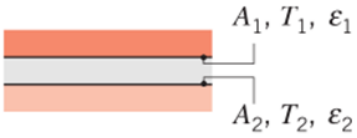
Innite Plane and Row of Cylinders



$$F_{ij} = 1 - \left[1 - \left(\frac{D}{s}\right)^2\right]^{1/2} + \left(\frac{D}{s}\right)\tan^{-1}\left[\left(\frac{s^2 - D^2}{D^2}\right)^{1/2}\right]$$

Geometry	Relation
Aligned Parallel Rectangles **(Figure 13.4)** 	$\overline{X} = X/L,\ \overline{Y} = Y/L$ $F_{ij} = \frac{2}{\pi \overline{X}\,\overline{Y}} \left\{ \ln \left[\frac{(1+\overline{X}^2)(1+\overline{Y}^2)}{1+\overline{X}^2+\overline{Y}^2} \right]^{1/2} + \overline{X}(1+\overline{Y}^2)^{1/2} \tan^{-1} \frac{\overline{X}}{(1+\overline{Y}^2)^{1/2}} + \overline{Y}(1+\overline{X}^2)^{1/2} \tan^{-1} \frac{\overline{Y}}{(1+\overline{X}^2)^{1/2}} - \overline{X} \tan^{-1} \overline{X} - \overline{Y} \tan^{-1} \overline{Y} \right\}$
Coaxial Parallel Disks **(Figure 13.5)** 	$R_i = r_i/L,\ R_j = r_j/L$ $S = 1 + \frac{1+R_j^2}{R_i^2}$ $F_{ij} = \frac{1}{2}\{S - [S^2 - 4(r_j/r_i)^2]^{1/2}\}$
Perpendicular Rectangles with a Common Edge **(Figure 13.6)** 	$H = Z/X,\ W = Y/X$ $F_{ij} = \frac{1}{\pi W} \left(W \tan^{-1} \frac{1}{W} + H \tan^{-1} \frac{1}{H} - (H^2+W^2)^{1/2} \tan^{-1} \frac{1}{(H^2+W^2)^{1/2}} + \frac{1}{4} \ln \left\{ \frac{(1+W^2)(1+H^2)}{1+W^2+H^2} \left[\frac{W^2(1+W^2+H^2)}{(1+W^2)(W^2+H^2)} \right]^{W^2} \times \left[\frac{H^2(1+H^2+W^2)}{(1+H^2)(H^2+W^2)} \right]^{H^2} \right\} \right)$

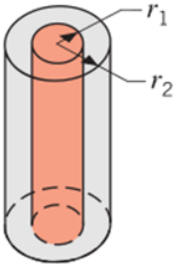
Large (Innite) Parallel Planes



$$A_1 = A_2 = A$$
$$F_{12} = 1$$

$$q_{12} = \frac{A\sigma(T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} - 1}$$

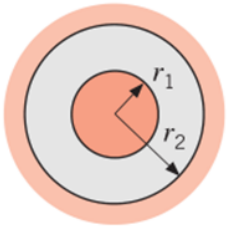
Long (Innite) Concentric Cylinders



$$\frac{A_1}{A_2} = \frac{r_1}{r_2}$$
$$F_{12} = 1$$

$$q_{12} = \frac{\sigma A_1(T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1} + \frac{1 - \varepsilon_2}{\varepsilon_2}\left(\frac{r_1}{r_2}\right)}$$

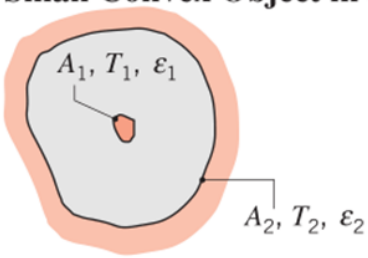
Concentric Spheres



$$\frac{A_1}{A_2} = \frac{r_1^2}{r_2^2}$$
$$F_{12} = 1$$

$$q_{12} = \frac{\sigma A_1(T_1^4 - T_2^4)}{\frac{1}{\varepsilon_1} + \frac{1 - \varepsilon_2}{\varepsilon_2}\left(\frac{r_1}{r_2}\right)^2}$$

Small Convex Object in a Large Cavity



$$\frac{A_1}{A_2} \approx 0$$
$$F_{12} = 1$$

$$q_{12} = \sigma A_1 \varepsilon_1 (T_1^4 - T_2^4)$$

Table 1: Contact Resistance for vacuum interfaces, $R''_{t,c} \times 10^4 \left(\frac{m^2 K}{W}\right)$

Material	$100\,(kPa)$	$10,000\,(kPa)$
Stainless Steel	6-25	0.7-4.0
Copper	1-10	0.1-0.5
Magnesium	1.5-3.5	0.2-0.4
Aluminum	1.5-5.0	0.2-0.4

Table 2: Contact Resistance for aluminum ($10\mu m$ surface roughness, $10kPa$ contact pressure)

Interfacial Fluid	$R''_{t,c} \times 10^4 \left(\frac{m^2 K}{W}\right)$
Air	2.75
Helium	1.05
Hydrogen	0.720
Silicone Oil	0.525
Glycerine	0.265

Table 3: Fin Tip Conditions

Case	Tip Condition	Temperature Distribution $\frac{\theta}{\theta_b}$	Fin Heat Transfer Rate q_f
A	Convection	$\frac{\cosh m(L-x)+\left(\frac{h}{mk}\right)\sinh m(L-x)}{\cosh mL+\left(\frac{h}{mK}\right)\sinh mL}$	$M\frac{\sinh mL+\left(\frac{h}{mk}\right)\cosh mL}{\cosh mL+\left(\frac{h}{mk}\right)\sinh mL}$
B	Adiabatic	$\frac{\cosh m(L-x)}{\cosh mL}$	$M\tanh mL$
C	Constant Temperature	$\frac{\left(\frac{\theta_L}{\theta_b}\right)\sinh mx+\sinh m(L-x)}{\sinh mL}$	$M\frac{\cosh mL-\frac{\theta_L}{\theta_b}}{\sinh mL}$
D	Infinite Fin ($L\to\infty$)	e^{-mx}	M

Table 4: Energy Balance Method Case Summary

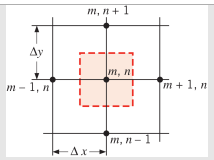
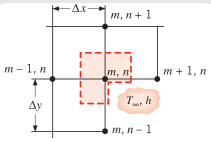
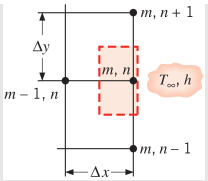
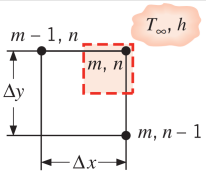
Case	Diagram	Equation
1		$T_{m,n+1} + T_{m,n-1} + T_{m+1,n} + T_{m-1,n} - 4T_{m,n} = 0$
2		$2\left(T_{m-1,n} - T_{m,n+1}\right) + \left(T_{m+1,n} + T_{m,n-1}\right) + 2\frac{h\Delta x}{k}T_\infty - 2\left(3 + \frac{h\Delta x}{k}\right)T_{m,n} = 0$
3		$\left(2T_{m-1,n} + T_{m,n+1} + T_{m,n-1}\right) + 2\frac{h\Delta x}{k}T_\infty - 2\left(2 + \frac{h\Delta x}{k}\right)T_{m,n} = 0$
4		$\left(T_{m,n-1} + T_{m-1,n}\right) + 2\frac{h\Delta x}{k}T_\infty - 2\left(1 + \frac{h\Delta x}{k}\right)T_{m,n} = 0$

Table 5: Cylinder In Cross Flow

Re_D	C	m
$0.4 - 4$	0.989	0.330
$4 - 40$	0.911	0.385
$40 - 4000$	0.683	0.466
$4000 - 40,000$	0.193	0.618
$40,000 - 400,000$	0.027	0.805

Table 6: Various Geometries In Cross Flow

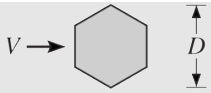
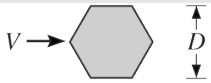
Geometry		Re_D	C	m
		$6000 - 60,000$	0.304	0.59
		$5000 - 60,000$	0.158	0.66
		$5200 - 20,400$ $20,400 - 105,000$	0.164 0.039	0.638 0.78
		$4500 - 90,700$	0.150	0.638
	Front Back	$10,000 - 50,000$ $7,000 - 80,000$	0.667 0.191	0.500 0.667

Table 7: Noncircular Tubes

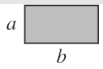
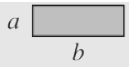
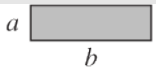
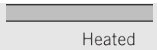
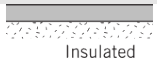
Cross Section	$\frac{b}{a}$	Uniform q_s''	Uniform T_s	fRe_{D_h}
		4.36	3.66	64
	1.0	3.61	2.98	57
	1.43	3.73	3.08	59
	2.0	4.12	3.39	62
	3.0	4.79	3.96	69
	4.0	5.33	4.44	73
	∞	8.23	7.54	96
	∞	5.39	4.86	96
		3.11	2.49	53

Table 8: Fouling Factors

Fluid	$R_f^{''}\left(m^2K/W\right)$
Seawater and treated boiler feedwater (below $50°C$)	0.001
Seawater and treated boiler feedwater (above $50°C$)	0.002
River water (below $50°C$)	0.0002 - 0.001
Fuel Oil	0.0009
Refrigerating Liquids	0.0002
Steam (nonoil bearing)	0.0001

Table 9: Blackbody Radiation Functions

$\lambda T\,(\mu mK)$	$F_{(0\to\lambda)}$	$I_{\lambda,b}(\lambda,T)/\sigma T^5\,(\mu mKsr)^{-1}$	$\frac{I_{\lambda,b}(\lambda,T)}{I_{\lambda,b}(\lambda_{max},T)}$
200	0.000000	$0.375034E-27$	0.000000
400	0.000000	$0.490335E-13$	0.000000
600	0.000000	$0.104046E-8$	0.000014
800	0.000016	$0.991126E-7$	0.001372
1000	0.000321	$0.118505E-5$	0.016406
1200	0.002134	$0.523927E-5$	0.072534
1400	0.007790	$0.134411E-4$	0.186082
1600	0.019718	$0.249130E-4$	0.344904
1800	0.039341	$0.375568E-4$	0.519949
2000	0.066728	$0.493432E-4$	0.683123
2200	0.100888	$0.589649E-4$	0.816329
2400	0.140256	$0.658866E-4$	0.912155
2600	0.183120	$0.701292E-4$	0.970891
2898	0.250108	$0.722318E-4$	1.000000
3000	0.273232	$0.720254E-4$	0.997143
3400	0.361735	$0.681544E-4$	0.943551
3800	0.443382	$0.615225E-4$	0.851737
4200	0.516014	$0.540394E-4$	0.748139
4600	0.579280	$0.467343E-4$	0.647004
5000	0.633747	$0.400813E-4$	0.554898
5400	0.680360	$0.342445E-4$	0.474092
5800	0.720158	$0.292301E-4$	0.404671
6200	0.754140	$0.249723E-4$	0.345724
6600	0.783199	$0.213786E-4$	0.295973
7000	0.808109	$0.183534E-4$	0.254090
7400	0.829527	$0.158073E-4$	0.218842
7800	0.848005	$0.136621E-4$	0.189143
8000	0.856288	$0.127185E-4$	0.176079
9000	0.890029	$0.901463E-5$	0.124801
10000	0.914199	$0.653279E-5$	0.090442
20000	0.985602	$0.623310E-6$	0.008629
30000	0.995340	$0.140469E-6$	0.001945
40000	0.997967	$0.473891E-7$	0.000656
50000	0.998953	$0.201605E-7$	0.000279
75000	0.999713	$0.418597E-8$	0.000058
100000	0.999905	$0.135752E-8$	0.000019